QUESTION 1

Why was C originally developed as a programming language?

Answer is Correct **Score: 1 / 1**

Options

A. To teach programming concepts

B. To support mobile app development

Correct

C. To develop operating systems like Unix

D. To build web applications

FEEDBACK

Solution: C was introduced to help develop operating systems, and it evolved alongside the Unix operating system, offering close hardware interaction.
Link: https://youtu.be/6KVgKWnGqJA?feature=shared&t=95

QUESTION 2

Which of the following are valid reasons C is still used in high-performance areas?

Answer is Correct **Score: 1 / 1**

Options

- [ ] A. It supports AI models
- [x] B. It offers fine-grained control over resources — Correct
- [x] C. It runs faster than many higher-level languages — Correct
- [ ] D. It simplifies web development

FEEDBACK

Solution: C remains popular in high-performance computing because it allows low-level hardware control and fast execution.
Link: https://youtu.be/6KVgKWnGqJA?feature=shared&t=301

QUESTION 3

Why is the stdint.h header important in embedded C?

Answer is Correct **Score: 1 / 1**

Options

A. It enables multitasking

B. It provides hardware drivers

Correct

C. It defines fixed-width data types like uint8_t, int16_t

D. It supports wireless communication

FEEDBACK

Solution: stdint.h defines precise-width integer types, which are crucial in embedded systems where memory and bit-size control is important.

Link: https://youtu.be/6KVgKWnGqJA?feature=shared&t=500

QUESTION 4

What is one primary benefit of using functions in C?

Answer is Correct **Score: 1 / 1**

Options

A. They reduce memory usage automatically

B. They eliminate the need for loops

Correct

C. They support modular design and code reuse

D. They replace global variables

FEEDBACK

Solution: Functions help in modularizing code, making it reusable, readable, and easier to maintain.

Link: https://youtu.be/6KVgKWnGqJA?feature=shared&t=991

QUESTION 5

Which factors are typical constraints in embedded system design?

Answer is Correct **Score: 1 / 1**

Options

Correct

- [x] A. Power consumption

- [ ] B. Screen resolution

Correct

- [x] C. Physical size

Correct

- [x] D. Cost

FEEDBACK

Solution: Embedded systems are built under constraints like limited power (often battery-based), compact size, and low cost.

Link: https://youtu.be/6KVgKWnGqJA?feature=shared&t=1491